Context compaction with one-shot summaries

An exploratory comparison on 100 HotpotQA and 100 QASPER binary questions

HotpotQA: LLM summary has the lowest compressed-condition Brier score (0.156; Full 0.159).

QASPER: LLM summary has the lowest compressed-condition Brier score (0.144; Full 0.163).

Reader: Qwen3-30B-A3B-Instruct-2507. Summary writer: DeepSeek-V3.2. Temperature 0; thinking off. Probabilities are verbalized p(Yes), not calibrated or token-logprob-derived posteriors.

The source-specific requested length is half its Qwen token count. Local methods enforce this cap; summary keeps its original first draft even if longer. There is no minimum retention requirement. No query, label or prediction is passed to a compressor; no cross-method minimum matching or post-truncation is used.

HotpotQA

Method	Accuracy	Brier	Log loss	Retained	Min-max	Over 50%
Full	81.0%	0.159	0.998	100.0%	100%-100%	-
No context	75.0%	0.186	1.804	0.0%	0%-0%	-
Selective Context	74.0%	0.190	0.569	48.8%	47%-50%	0/100
LLMLingua-2	77.0%	0.187	0.835	49.0%	48%-50%	0/100
LLM summary	80.0%	0.156	0.969	50.9%	23%-80%	51/100
DAC	74.0%	0.188	0.811	48.3%	48%-49%	0/100

QASPER

Method	Accuracy	Brier	Log loss	Retained	Min-max	Over 50%
Full	80.0%	0.163	0.493	100.0%	100%-100%	-
No context	71.0%	0.250	0.693	0.0%	0%-0%	-
Selective Context	78.0%	0.167	0.504	48.8%	48%-50%	0/100
LLMLingua-2	77.0%	0.167	0.505	49.0%	48%-50%	0/100
LLM summary	79.0%	0.144	0.432	14.6%	6%-39%	0/100
DAC	79.0%	0.160	0.489	48.5%	48%-49%	0/100

Accuracy: higher is better. Brier and log loss: lower is better. Retention uses pooled token counts. Log loss clips only for the logarithm at epsilon=1e-12.

Local methods respect a 50% cap. Summary keeps the original first draft even when over target: HotpotQA has 51/100 exceedances, QASPER 0/100. Actual lengths differ, so this is not an equal-token or common-hard-cap ranking.

Bidirectional InfoGap

Each cell is G(row, column): the largest normalized decision-payoff advantage of the row predictor over the column predictor. Darker cells show larger gaps. Both panels use the same scale.

HotpotQA

Row / column	Full	No ctx	SC	LLMLingua-2	Summary	DAC
Full	0.000	0.087	0.082	0.057	0.035	0.093
No ctx	0.063	0.000	0.106	0.100	0.035	0.046
SC	0.069	0.100	0.000	0.072	0.040	0.057
LLMLingua-2	0.020	0.080	0.030	0.000	0.020	0.064
Summary	0.092	0.070	0.079	0.097	0.000	0.065
DAC	0.089	0.080	0.062	0.092	0.033	0.000

QASPER

Row / column	Full	No ctx	SC	LLMLingua-2	Summary	DAC
Full	0.000	0.510	0.050	0.090	0.043	0.034
No ctx	0.071	0.000	0.060	0.077	0.001	0.088
SC	0.024	0.490	0.000	0.040	0.043	0.028
LLMLingua-2	0.046	0.480	0.059	0.000	0.030	0.019
Summary	0.130	0.500	0.117	0.100	0.000	0.101
DAC	0.036	0.500	0.040	0.057	0.043	0.000

G(A,B) = 2 sup_t mean[(t-y){1[p_A <= t] - 1[p_B <= t]}]. All observed probability breakpoints and one-sided limits are scanned. The unscaled value is also saved.

Two positive directions identify decision tradeoffs, not a net information gain. A positive G(compressed, Full) alone does not establish Blackwell dominance.

Uncertainty, validity and reproducibility

Uncertainty relative to Full

Dataset / method	G(Full, method), 95% interval	G(method, Full), 95% interval
HotpotQA / Selective Context	0.082 [0.021, 0.179]	0.069 [0.026, 0.161]
HotpotQA / LLMLingua-2	0.057 [0.011, 0.157]	0.020 [0.000, 0.100]
HotpotQA / LLM summary	0.035 [0.006, 0.110]	0.092 [0.015, 0.199]
HotpotQA / DAC	0.093 [0.042, 0.188]	0.089 [0.030, 0.200]
QASPER / Selective Context	0.050 [0.015, 0.112]	0.024 [0.005, 0.099]
QASPER / LLMLingua-2	0.090 [0.034, 0.166]	0.046 [0.011, 0.108]
QASPER / LLM summary	0.043 [0.003, 0.111]	0.130 [0.052, 0.222]
QASPER / DAC	0.034 [0.021, 0.099]	0.036 [0.007, 0.115]

Intervals: 500 paired cluster-bootstrap draws by source context; descriptive percentile intervals, not dominance tests. Thresholds are selected on the same evaluation examples.

Sampling

Seed 20261025; uniform draws from previously defined pools, without model-outcome selection. HotpotQA: 45 Yes / 55 No, distractor dev. QASPER: 71 Yes / 29 No, 97 papers, consistent binary annotations with text-verifiable evidence, train split. Evidence filters define the subset but do not select compressor input. This is not official full-benchmark scoring.

Prompt controls

Reader prompt and generation settings are unchanged across conditions. All 197 original first drafts share one summary prompt template, without output-ceiling retries or length revisions. All compression is query-blind. Historical requests are reused; other five conditions and reader parameters stay unchanged.

DAC implementation

Pinned official code; Qwen2-0.5B-Instruct, float32, dynamic_attn_ppl, additive fusion, alpha=0.8, 10 iterations. Query-row blocks retain attention to the entire causal prefix; vocabulary logits are also blocked. Two short examples produced identical reference outputs. The full longest source (16,755 tokens) passed at a 3.25 GiB peak. Floating-point reduction differences remain possible.

Audit and interpretation

Summaries reuse the original first complete generations; no new summary call is made. The completed cap-only study is preserved separately. All other five conditions are unchanged. All 1,200 probabilities replay from exact request caches; all 72 matrix cells were independently checked. Historical results remain unchanged. Cached predictions span earlier collection dates. This evaluates reader-dependent usefulness on a fixed query distribution, not intrinsic query-agnostic informativeness. Calibration and small-sample uncertainty limit dominance claims.

Cost

Conservative cumulative accounting including earlier revisions and unresolved-usage reserves: CNY 8.28, within the CNY 20 allowance. This is not the provider invoice; local GPU compute and reused caches create no new model API charge.

Sources: InfoGap; DAC (ACL 2025); LLMLingua-2; Selective Context. The companion HTML includes an offline per-example audit viewer.